

OpenEd Users, Roles, and IAM Document

Scope

OpenEd v1 has exactly three user categories:

1. OpenEd Team.
2. Students / Learners.
3. Educators.

This document defines v1 user roles, permissions, onboarding states, and IAM boundaries. It intentionally excludes employers, universities, organization admins, paid cohorts, and enterprise portals until later phases.

Role model

OpenEd Team

The OpenEd Team manages platform quality and trust.

Capabilities:

- Review courses submitted by educators.
- Approve, reject, or request changes.
- View course QA reports.
- Manage reported content.
- Manage public/private course visibility.
- Manage educator verification status.
- View aggregate analytics.
- Manage platform policies.
- Manage role assignments.
- Moderate harmful or spam content.

Restrictions:

- Cannot access learner BYOK keys.
- Should not access private learner artifacts except for support/moderation with audit logs.

- Cannot impersonate users without explicit audited support flow.

Student / Learner

The learner uses OpenEd to learn, ask, practice, and build proof.

Capabilities:

- Create account and profile.
- Enroll in free courses.
- Save lesson progress.
- Use AI tutor if configured.
- Manage BYOK settings.
- Complete quizzes.
- Submit artifacts.
- Receive feedback.
- Build SkillProof portfolio.
- Export/share portfolio items.
- Delete own data where legally and technically feasible.

Restrictions:

- Cannot edit course content.
- Cannot view other learners' private artifacts.
- Cannot publish courses unless upgraded to educator role.

Educator

The educator creates courses.

Capabilities:

- Create course drafts.
- Add lessons, sources, media, quizzes, rubrics, artifacts.
- Preview course as learner.
- Run course QA checks.
- Submit course for review.
- Edit own draft courses.
- View own course analytics.
- Respond to learner feedback if enabled.
- Review learner artifacts only for courses they own and only when that feature is enabled.

Restrictions:

- Cannot publish without OpenEd Team approval.
- Cannot access learner BYOK keys.
- Cannot override platform safety policy.
- Cannot view analytics across other educators' courses.

Permission matrix

Capability	Learner	Educator	OpenEd Team
View public courses	Yes	Yes	Yes
Enroll in course	Yes	Yes	Yes
Save progress	Yes	Yes	Optional
Use AI tutor	Yes	Yes	Yes
Create course draft	No	Yes	Yes
Submit course for QA	No	Yes	Yes
Approve course	No	No	Yes
View course QA	Own learning only	Own courses	All
Submit artifacts	Yes	Optional	Optional
Grade artifacts	No	Own courses	Review mode
Moderate content	No	Own course comments only	Yes
Manage roles	No	No	Yes
View platform analytics	No	Own courses	Aggregate/all
Access BYOK key	User only	User only	Never

Authentication

V1 should support:

- Email/password sign-up and login.
- Forgot password.
- Email verification if enabled.
- Session persistence.
- Protected routes.

- Profile creation on sign-up.
- Role-based route guards.

Later:

- Google login.
- Magic links.
- Passkeys.
- MFA.
- Institution SSO.

Supabase security posture

Use Supabase Auth and Postgres Row Level Security. RLS should be enabled on all public schema tables that contain user or course data. Policies should use `auth.uid()` for ownership checks and explicitly check authentication. The service role key must never appear in frontend code.

Core tables

Table	Purpose
profiles	User profile, display name, role, onboarding status.
courses	Course metadata and status.
course_versions	Versioned course content.
course_enrollments	Learner-course relationships.
lesson_progress	Per-lesson progress and completion.
artifacts	Learner-created proof objects.
artifact_feedback	Rubric feedback and revisions.
tutor_sessions	Non-sensitive tutor session metadata.
educator_courses	Educator-course ownership.
course_reviews	OpenEd Team QA decisions.
content_reports	Abuse/safety reports.
audit_events	Sensitive admin and moderation actions.

Route access

Public

- Landing.
- Course preview.
- Login.
- Signup.
- Forgot password.
- Public portfolio items if learner explicitly shares.

Learner-protected

- Dashboard.
- Learning map.
- Lesson workspace.
- Quiz.
- Artifact builder.
- Portfolio.
- BYOK settings.

Educator-protected

- Educator dashboard.
- Course drafts.
- Lesson builder.
- Source card editor.
- Quiz/rubric builder.
- Publish-readiness QA.

OpenEd Team-protected

- Course review queue.
- Reports/moderation.
- Platform analytics.
- Role management.
- Policy configuration.

IAM failure modes

Failure mode	Prevention
Learner sees another learner's artifact	RLS owner checks, route guards, test fixtures.
Educator publishes unreviewed course	Course status machine; publish action restricted.
Admin actions not auditable	audit_events table required.
BYOK key exposed	Never store on server; never log; optional browser storage warning.
Role escalation bug	Server-side checks and RLS, not UI-only checks.
Deleted course breaks learner records	Soft-delete and versioned course references.

V1 role-state machine

```
anonymous -> learner_pending_email -> learner_active
learner_active -> educator_applicant -> educator_active
educator_active -> educator_suspended
open_ed_team_member -> open_ed_reviewer -> open_ed_admin
```

Educator role should not be self-granted in v1 unless the platform explicitly allows public course drafting. Publishing rights always require OpenEd Team approval.

Source notes

- This document is grounded in the current OpenEd/FME project context and the following reference set where applicable:
- Uploaded Frontier Model Evaluations prototype code bundle, latest inspected bundle: Design System for Frontier Model Evaluation (5).zip.
 - Uploaded Frontier Model Evaluations 51-hour curriculum and Module 1 foundations files.
 - Coursera, Udemy, DataCamp, Moodle, Khanmigo, Duolingo public documentation and product pages.
 - Supabase Auth and Row Level Security documentation.
 - OpenAI API key safety guidance, MDN Web APIs, and YouTube IFrame Player API documentation.
 - Bloom's 2 sigma problem, intelligent tutoring systems research, CourseAssist, LearnLM-Tutor, DCCI, and UNESCO guidance for generative AI in education.

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